

MCHA4400

Vision-based navigation

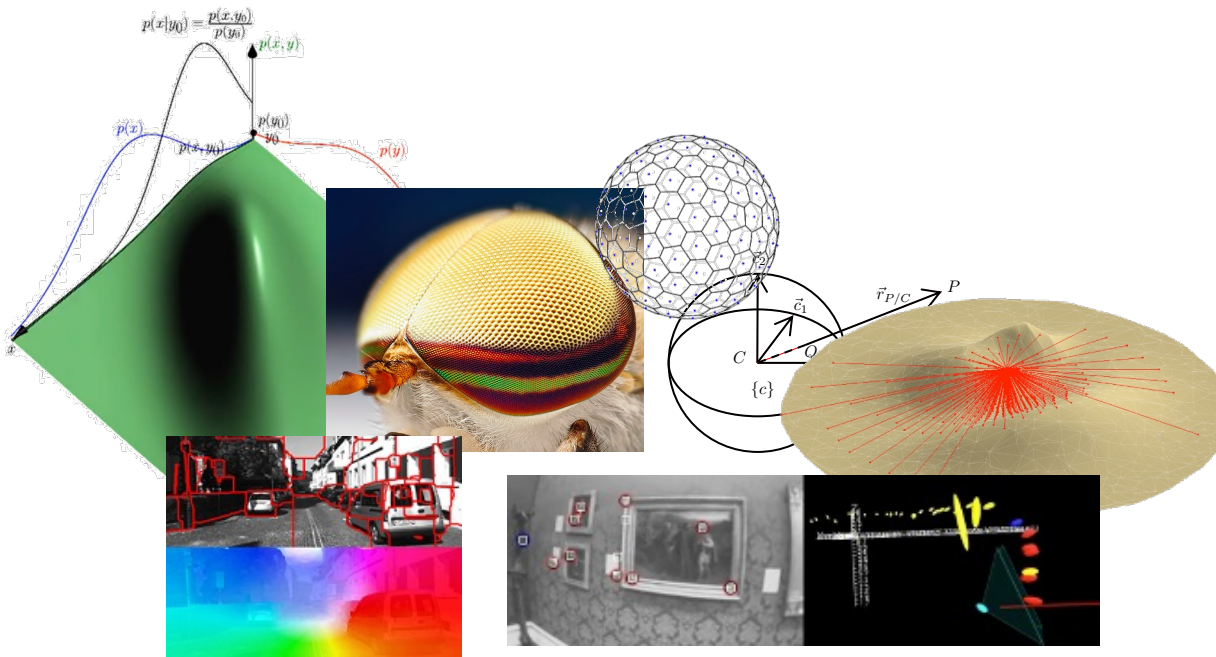


THE UNIVERSITY OF
NEWCASTLE
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FACULTY OF
ENGINEERING AND
BUILT ENVIRONMENT



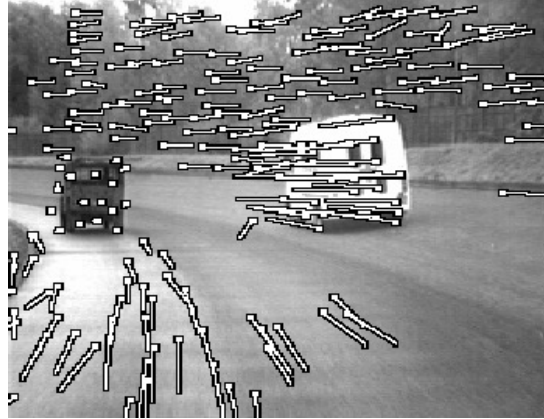
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Image motion and optic flow

Image flow



Sparse

- Attempt to identify common point features between consecutive image frames
- Flow field may be globally inconsistent
- Requires high-texture environments

Dense

- Attempt to correlate image regions between consecutive frames
- Pixel density globally consistent flow field (assumptions)
- Computationally expensive

Some algorithms available in OpenCV

- Lucas-Kanade (sparse)
- Farneback (dense)

Sparse flow



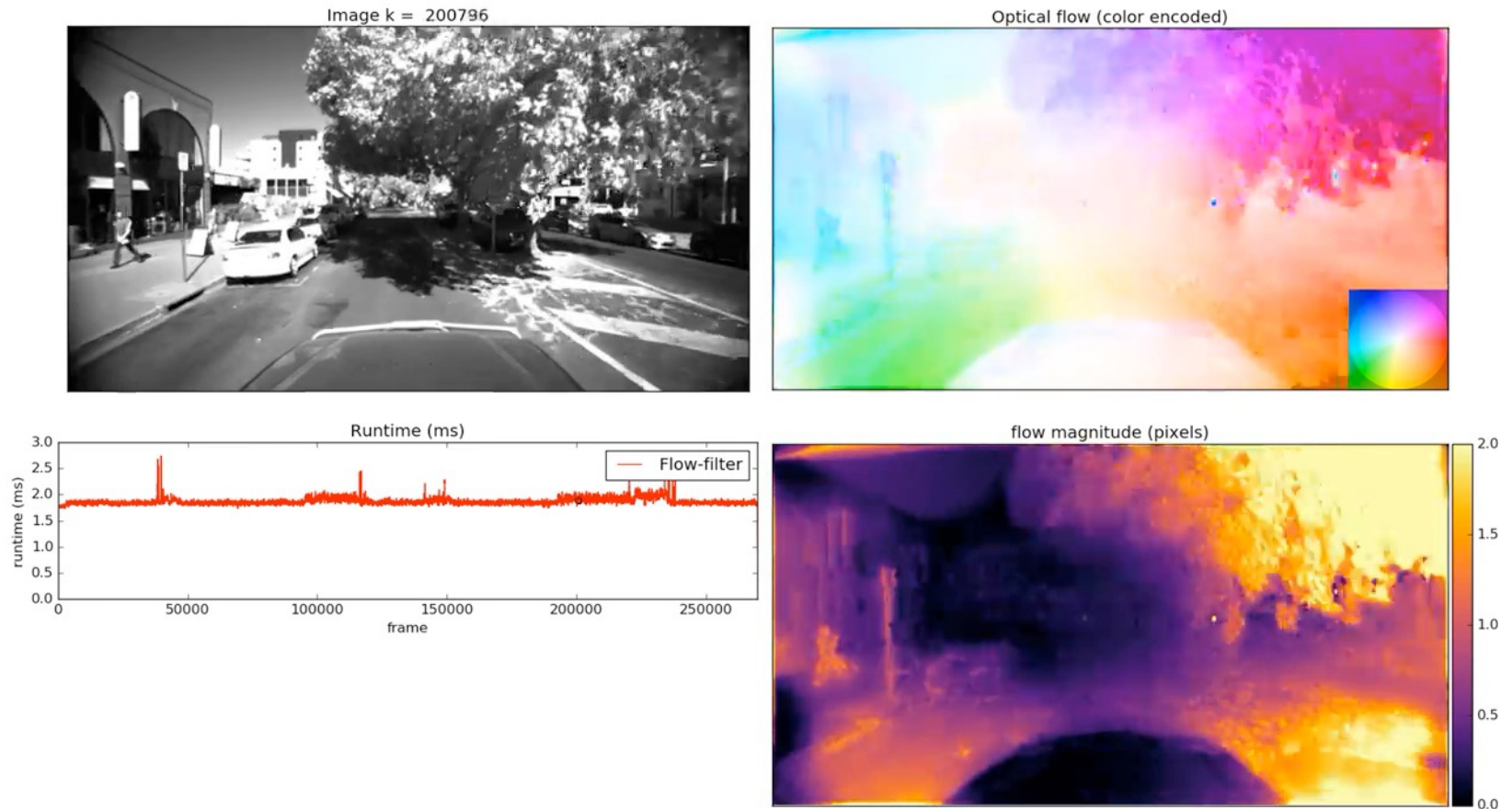
Dense flow

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https://youtu.be/5rR_9YIcg_s

Real-time dense optic flow



https://youtu.be/_Rpi7WS-HTw

Projective geometry

The projective space $\mathbb{P}^n$

The **real projective space of dimension n** can be defined as

$$\mathbb{P}^n = \{ \mathbf{x} \in \mathbb{R}^{n+1} \setminus \{ \mathbf{0} \} \mid \mathbf{x} \sim \lambda \mathbf{x}, \forall \lambda \neq 0 \}$$

Elements of $\mathbb{P}^n$ are considered equivalent up to scale and are called **homogeneous coordinates**.

For example,

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \sim \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \in \mathbb{P}^2$$

are considered the same element in $\mathbb{P}^2$.

The projective space $\mathbb{P}^n$

In robotics, we typically concatenate an element of $\mathbb{R}^n$ with 1 to form a homogeneous vector in $\mathbb{P}^n$.

For example, let $\mathbf{r} \in \mathbb{R}^3$. Then,

$$\mathbf{p} = \begin{bmatrix} \mathbf{r} \\ 1 \end{bmatrix} \in \mathbb{P}^3$$

A kinematic relationship using homogeneous coordinates is unaffected by scaling. For example, for some $\kappa \neq 0$, then

$$\begin{aligned} \kappa \begin{bmatrix} \mathbf{r}_{P/A}^a \\ 1 \end{bmatrix} &= \kappa \begin{bmatrix} \mathbf{R}_b^a & \mathbf{r}_{B/A}^a \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{r}_{P/B}^b \\ 1 \end{bmatrix} \\ \underbrace{\begin{bmatrix} \kappa \mathbf{r}_{P/A}^a \\ \kappa \end{bmatrix}}_{\mathbf{p}_{P/A}^a} &= \underbrace{\begin{bmatrix} \mathbf{R}_b^a & \mathbf{r}_{B/A}^a \\ \mathbf{0} & 1 \end{bmatrix}}_{\mathbf{T}_b^a} \underbrace{\begin{bmatrix} \kappa \mathbf{r}_{P/B}^b \\ \kappa \end{bmatrix}}_{\mathbf{p}_{P/B}^b} \end{aligned}$$

The projective space $\mathbb{P}^n$

Projective spaces can be used to greatly simplify some operations.

- In robotics, the equations for the transformation of a vector are affine when expressed in non-homogeneous coordinates ($\mathbb{R}^3 \rightarrow \mathbb{R}^3$), but are linear in homogeneous coordinates ($\mathbb{P}^3 \rightarrow \mathbb{P}^3$).
- In vision, the equations for perspective projection are non-linear when expressed in non-homogeneous coordinates ($\mathbb{R}^3 \rightarrow \mathbb{R}^2$), but are linear in homogeneous coordinates ($\mathbb{P}^3 \rightarrow \mathbb{P}^2$).

The projective plane $\mathbb{P}^2$

The non-homogeneous coordinates of a point in the image plane is given by a 2-vector, while the homogeneous coordinates are a 3-vector.

The simplest way to map $\mathbb{R}^2 \rightarrow \mathbb{P}^2$ is to set the third component to one, i.e.,

$$\begin{bmatrix} u \\ v \end{bmatrix} \mapsto \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \sim \begin{bmatrix} \lambda u \\ \lambda v \\ \lambda \end{bmatrix}, \quad \forall \lambda \neq 0$$

To map $\mathbb{P}^2 \rightarrow \mathbb{R}^2$, we divide the first two components by the third, i.e.,

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} \mapsto \begin{bmatrix} \frac{u}{w} \\ \frac{v}{w} \end{bmatrix}$$

The projective plane $\mathbb{P}^2$

We can also represent lines in the image plane in homogeneous coordinates.

Let $\mathbf{l}$ be a line and $\mathbf{p}$ be a point in homogeneous coordinates,

$$\mathbf{l} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \quad \mathbf{p} = \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

The points $\mathbf{p}$ that lie on the line $\mathbf{l}$ satisfy

$$\mathbf{l}^T \mathbf{p} = 0$$

$$au + bv + cw = 0$$

If we divide by w and let $x = u/w$, $y = v/w$, we recover the standard equation for a line

$$ax + by + c = 0$$

Points and lines at infinity

Homogeneous points of the form

$$\mathbf{p} = \begin{bmatrix} u \\ v \\ 0 \end{bmatrix} \in \mathbb{P}^2$$

don't map to any real point. This is a *point at infinity* in the direction defined by u, v .

All such points lie on the homogeneous line,

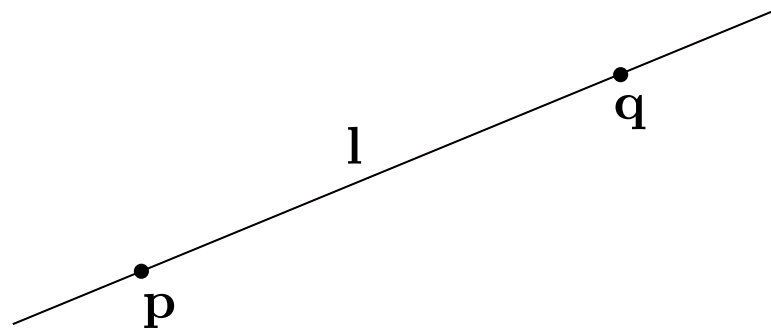
$$\mathbf{l} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \in \mathbb{P}^2$$

which doesn't map to any real line. This is the *line at infinity*, which can be verified from

$$\mathbf{l}^T \mathbf{p} = 0$$

The projective plane $\mathbb{P}^2$

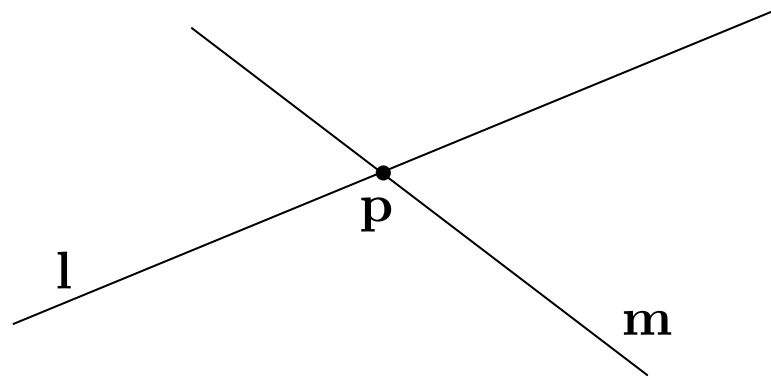
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Let $\mathbf{p}, \mathbf{q} \in \mathbb{P}^2$ be homogeneous points. Then, the homogeneous line $\mathbf{l} \in \mathbb{P}^2$ that passes through $\mathbf{p}$ and $\mathbf{q}$ is given by

$$\mathbf{l} = \mathbf{p} \times \mathbf{q} = \mathbf{S}(\mathbf{p})\mathbf{q}$$

The projective plane $\mathbb{P}^2$



Let $l, m \in \mathbb{P}^2$ be homogeneous lines. Then, the homogeneous point $p \in \mathbb{P}^2$ that lies on both l and m is given by

$$p = l \times m = S(l)m$$

The projective plane $\mathbb{P}^2$

Example: Find the intersection of the lines $x = 1$ and $y = 1$.

$$x = 1 \implies -1x + 0y + 1 = 0 \implies \mathbf{l} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \in \mathbb{P}^2$$

$$y = 1 \implies 0x - 1y + 1 = 0 \implies \mathbf{m} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \in \mathbb{P}^2$$

$$\mathbf{p} = \mathbf{l} \times \mathbf{m} = \mathbf{S}(\mathbf{l})\mathbf{m}$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \in \mathbb{P}^2 \mapsto \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \in \mathbb{R}^2$$

which is the point $x = 1$ and $y = 1$ as expected.

The projective plane $\mathbb{P}^2$

Example: Find the intersection of the lines $x = 1$ and $x = 2$.

$$x = 1 \implies -1x + 0y + 1 = 0 \implies \mathbf{l} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \in \mathbb{P}^2$$

$$x = 2 \implies -1x + 0y + 2 = 0 \implies \mathbf{m} = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} \in \mathbb{P}^2$$

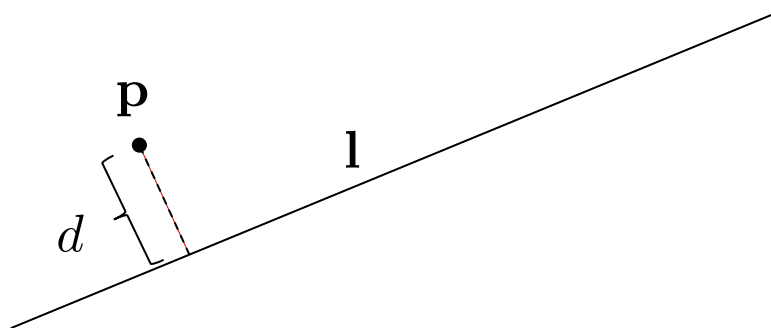
$$\mathbf{p} = \mathbf{l} \times \mathbf{m} = \mathbf{S}(\mathbf{l})\mathbf{m}$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \in \mathbb{P}^2$$

which is a point at infinity in the direction of the y -axis.

The projective plane $\mathbb{P}^2$

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Let $\mathbf{p} \in \mathbb{P}^2$ be a homogeneous point and let $\mathbf{l} \in \mathbb{P}^2$ be a homogeneous line. Then, the perpendicular distance between $\mathbf{p}$ and $\mathbf{l}$ is given by

$$d = \mathbf{n}_l(\mathbf{l})^\top \mathbf{n}_p(\mathbf{p})$$

where we *normalise* the point and line as follows:

$$\mathbf{n}_p \left(\begin{bmatrix} u \\ v \\ w \end{bmatrix} \right) = \begin{bmatrix} \frac{u}{w} \\ \frac{v}{w} \\ 1 \end{bmatrix} \quad \mathbf{n}_l \left(\begin{bmatrix} a \\ b \\ c \end{bmatrix} \right) = \frac{1}{\sqrt{a^2 + b^2}} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

The projective space $\mathbb{P}^3$

The non-homogeneous coordinates of a point in space is given by a 3-vector, while the homogeneous coordinates are a 4-vector.

The simplest way to map $\mathbb{R}^3 \rightarrow \mathbb{P}^3$ is to set the fourth component to one, i.e.,

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \mapsto \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \sim \begin{bmatrix} \lambda x \\ \lambda y \\ \lambda z \\ \lambda \end{bmatrix}, \quad \forall \lambda \neq 0$$

To map $\mathbb{P}^3 \rightarrow \mathbb{R}^3$, we divide the first three components by the fourth, i.e.,

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \mapsto \begin{bmatrix} \frac{x}{w} \\ \frac{y}{w} \\ \frac{z}{w} \end{bmatrix}$$

The projective space $\mathbb{P}^3$

We can also represent planes in space in homogeneous coordinates.

Let π be a plane and $\mathbf{x}$ be a point in homogeneous coordinates,

$$\pi = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$$

The points $\mathbf{x}$ that lie on the plane π satisfy

$$\pi^T \mathbf{x} = 0$$

$$ax + by + cz + dw = 0$$

If we divide by w , we recover an equation for a plane

$$\frac{a}{w}x + \frac{b}{w}y + \frac{c}{w}z + d = 0$$

Points and planes at infinity

Homogeneous points of the form

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \\ 0 \end{bmatrix} \in \mathbb{P}^3$$

don't map to any real point. This is a *point at infinity* in the direction defined by x, y, z .

All such points lie on the homogeneous plane,

$$\pi = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \in \mathbb{P}^3$$

which doesn't map to any real plane. This is the *plane at infinity*, which can be verified from

$$\pi^T \mathbf{x} = 0$$

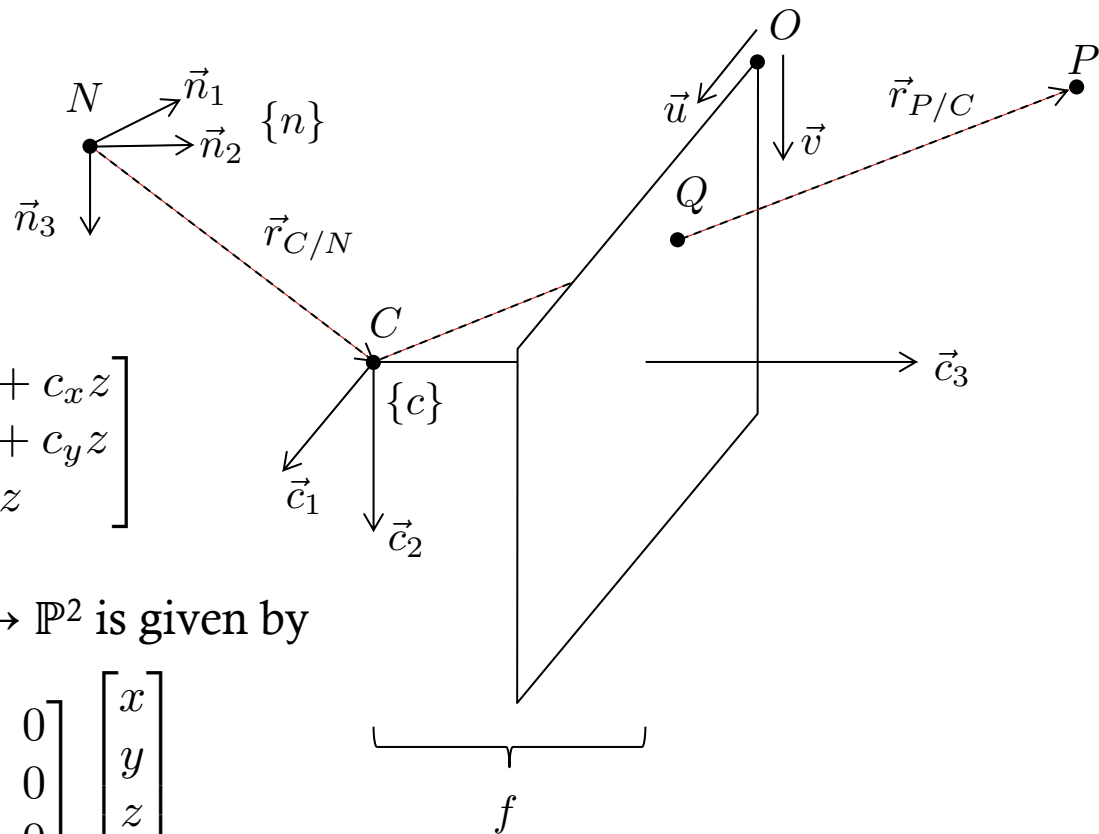
Planar camera model

Let $\mathbf{p}_{Q/O}^i \in \mathbb{P}^2$ be the homogeneous coordinates of Q w.r.t O in the image plane.

$$\mathbf{p}_{Q/O}^i = \begin{bmatrix} \frac{f_x}{z}x + c_x \\ \frac{f_y}{z}y + c_y \\ 1 \end{bmatrix} = \begin{bmatrix} f_x x + c_x z \\ f_y y + c_y z \\ z \end{bmatrix}$$

Then, the projection from $\mathbb{P}^3 \rightarrow \mathbb{P}^2$ is given by

$$\underbrace{\begin{bmatrix} f_x x + c_x z \\ f_y y + c_y z \\ z \end{bmatrix}}_{\mathbf{p}_{Q/O}^i} = \underbrace{\begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\bar{\mathbf{K}}} \underbrace{\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}}_{\mathbf{p}_{P/C}^c}$$



Planar camera model

For a camera with no lens distortion, we can write the camera model as a mapping from $\mathbb{P}^3 \rightarrow \mathbb{P}^2$,

$$\underbrace{\begin{bmatrix} f_x x + c_x z \\ f_y y + c_y z \\ z \end{bmatrix}}_{\mathbf{p}_{Q/O}^i} = \underbrace{\begin{bmatrix} f_x & 0 & c_x & 0 \\ 0 & f_y & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\bar{\mathbf{K}}} \underbrace{\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}}_{\mathbf{p}_{P/C}^c}$$

$$\mathbf{r}_{Q/O}^i = \begin{bmatrix} f_x \frac{x}{z} + c_x \\ f_y \frac{y}{z} + c_y \end{bmatrix}$$

$$\bar{\mathbf{K}} = [\mathbf{K} \quad \mathbf{0}]$$

$$\mathbf{p}_{Q/O}^i = \bar{\mathbf{K}} \mathbf{p}_{P/C}^c = \mathbf{K} \mathbf{r}_{P/C}^c$$

This compact form is convenient for performing kinematic transformations.

Using the **point alias** property of SE(3) yields

$$\mathbf{p}_{Q/O}^i = \bar{\mathbf{K}} \mathbf{T}_n^c \mathbf{p}_{P/N}^n$$

This is the homogeneous equivalent of the `worldToPixel` function (neglecting lens distortion).

Image of objects at infinity



Undistorted features

In OpenCV, the homogeneous camera model takes the form

$$\tilde{\mathbf{p}}_{Q/O}^i = \mathbf{K}\boldsymbol{\varphi}(\mathbf{r}_{P/C}^c)$$

where $\boldsymbol{\varphi}(\cdot)$ is the lens distortion function.

This can be rearranged to find (up to scale),

$$\begin{aligned}\boldsymbol{\varphi}(\mathbf{r}_{P/C}^c) &= \mathbf{K}^{-1}\tilde{\mathbf{p}}_{Q/O}^i \\ \mathbf{r}_{P/C}^c &= \boldsymbol{\varphi}^{-1}(\mathbf{K}^{-1}\tilde{\mathbf{p}}_{Q/O}^i)\end{aligned}$$

Let $\mathbf{p} \in \mathbb{P}^2$ be the homogeneous coordinates of an undistorted image feature,

$$\mathbf{p}_{Q/O}^i = \mathbf{K}\mathbf{r}_{P/C}^c$$

Then, we can express the undistorted features in terms of the measured features as follows:

$$\mathbf{p}_{Q/O}^i = \mathbf{K}\boldsymbol{\varphi}^{-1}(\mathbf{K}^{-1}\tilde{\mathbf{p}}_{Q/O}^i)$$

Epipolar geometry

Consider two views of the point P .

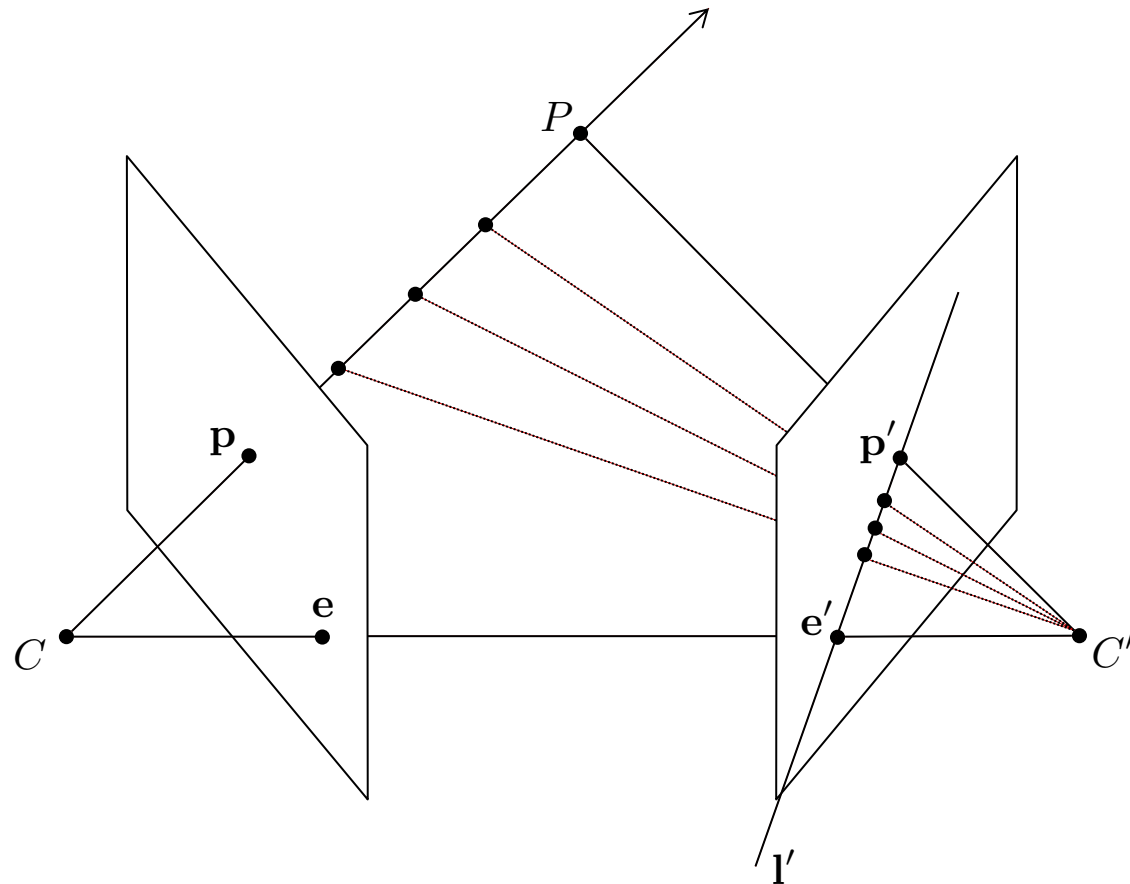
Let C and C' be the camera optical centres.

Let $\mathbf{p}, \mathbf{p}' \in \mathbb{P}^2$ be the homogeneous coordinates of the point in each view.

The **epipole** is the line joining the camera centres, through $\mathbf{e}, \mathbf{e}' \in \mathbb{P}^2$.

The **epipolar plane** is formed by the points C, C' and P .

An **epipolar line** is the intersection of the epipolar plane with an image plane, e.g., $\mathbf{l}' \in \mathbb{P}^2$.



Epipolar geometry

For brevity, we will drop the adornments on projective points, and consider two views of P as follows:

$$\begin{aligned}\mathbf{p} &= \mathbf{p}_{Q/O}^i = \bar{\mathbf{K}} \mathbf{T}_n^c \mathbf{p}_{P/N}^n \\ \mathbf{p}' &= \mathbf{p}_{Q'/O'}^{i'} = \bar{\mathbf{K}} \mathbf{T}_n^{c'} \mathbf{p}_{P/N}^n\end{aligned}$$

We aim to relate $\mathbf{p}'$ to $\mathbf{p}$ via the common world point.

For a given $\mathbf{p}$, the first equation is satisfied for a one-parameter family of solutions that correspond to all the world points along the view ray.

$$\mathbf{p}_{P/N}^n = \mathbf{T}_c^n \begin{bmatrix} \mathbf{K}^{-1} \mathbf{p} \\ \lambda \end{bmatrix}$$

for some value of the parameter λ . This can be validated as follows:

$$\mathbf{p} = [\mathbf{K} \quad \mathbf{0}] \mathbf{T}_n^c \underbrace{\mathbf{T}_c^n \begin{bmatrix} \mathbf{K}^{-1} \mathbf{p} \\ \lambda \end{bmatrix}}_{\mathbf{p}_{P/N}^n}$$

Epipolar geometry

Substituting

$$\mathbf{p}_{P/N}^n = \mathbf{T}_c^n \begin{bmatrix} \mathbf{K}^{-1} \mathbf{p} \\ \lambda \end{bmatrix}$$

into the second equation yields

$$\begin{aligned} \mathbf{p}'(\lambda) &= \bar{\mathbf{K}} \mathbf{T}_n^{c'} \mathbf{T}_c^n \begin{bmatrix} \mathbf{K}^{-1} \mathbf{p} \\ \lambda \end{bmatrix} \\ &= \bar{\mathbf{K}} \mathbf{T}_c^{c'} \begin{bmatrix} \mathbf{K}^{-1} \mathbf{p} \\ \lambda \end{bmatrix} \\ &= [\mathbf{K} \quad \mathbf{0}] \begin{bmatrix} \mathbf{R}_c^{c'} & \mathbf{r}_{C/C'}^{c'} \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} \mathbf{K}^{-1} \mathbf{p} \\ \lambda \end{bmatrix} \\ &= \mathbf{K} \mathbf{R}_c^{c'} \mathbf{K}^{-1} \mathbf{p} + \lambda \mathbf{K} \mathbf{r}_{C/C'}^{c'} \end{aligned}$$

Epipolar geometry

$$\mathbf{p}'(\lambda) = \mathbf{K}\mathbf{R}_c^{c'}\mathbf{K}^{-1}\mathbf{p} + \lambda\mathbf{K}\mathbf{r}_{C/C'}^{c'}$$

Let's take two points of interest and use them to construct the epipolar line.

Take $\lambda = 0$ and $\lambda = \infty$,

$$\mathbf{p}'(0) = \mathbf{K}\mathbf{R}_c^{c'}\mathbf{K}^{-1}\mathbf{p}$$

$$\mathbf{p}'(\infty) = \mathbf{K}\mathbf{r}_{C/C'}^{c'}$$

Then, the epipolar line that joins these points is given by

$$\mathbf{l}' = \mathbf{p}'(\infty) \times \mathbf{p}'(0)$$

$$= \mathbf{S}(\mathbf{K}\mathbf{r}_{C/C'}^{c'})\mathbf{K}\mathbf{R}_c^{c'}\mathbf{K}^{-1}\mathbf{p}$$

$$= \mathbf{K}^{-\top}\mathbf{S}(\mathbf{r}_{C/C'}^{c'})\mathbf{K}^{-1}\mathbf{K}\mathbf{R}_c^{c'}\mathbf{K}^{-1}\mathbf{p}$$

$$= \mathbf{K}^{-\top}\mathbf{S}(\mathbf{r}_{C/C'}^{c'})\mathbf{R}_c^{c'}\mathbf{K}^{-1}\mathbf{p}$$

$$\mathbf{S}(\mathbf{B}\mathbf{a}) = (\det \mathbf{B})\mathbf{B}^{-\top}\mathbf{S}(\mathbf{a})\mathbf{B}^{-1}$$

Fundamental matrix

In homogeneous coordinates, the epipolar line $\mathbf{l}'$ in the second image that corresponds to a point $\mathbf{p}$ measured in the first image is given by the linear relationship

$$\mathbf{l}' = \mathbf{F}\mathbf{p}$$

where

$$\begin{aligned}\mathbf{F} &= \mathbf{K}^{-\top} \mathbf{S}(\mathbf{r}_{C/C'}^{c'}) \mathbf{R}_c^{c'} \mathbf{K}^{-1} = \mathbf{K}^{-\top} \mathbf{S}(\mathbf{R}_n^{c'} \mathbf{r}_{C/C'}^n) \mathbf{R}_c^{c'} \mathbf{K}^{-1} \\ &= \mathbf{K}^{-\top} \mathbf{S}(\mathbf{R}_n^{c'} (\mathbf{r}_{C/N}^n - \mathbf{r}_{C'/N}^n)) \mathbf{R}_c^{c'} \mathbf{K}^{-1} \\ &= \mathbf{K}^{-\top} \mathbf{R}_n^{c'} \mathbf{S}(\mathbf{r}_{C/N}^n - \mathbf{r}_{C'/N}^n) \mathbf{R}_c^n \mathbf{R}_c^{c'} \mathbf{K}^{-1} \\ \mathbf{F} &= \mathbf{K}^{-\top} (\mathbf{R}_c^n)^\top \mathbf{S}(\mathbf{r}_{C/N}^n - \mathbf{r}_{C'/N}^n) \mathbf{R}_c^n \mathbf{K}^{-1}\end{aligned}$$

is the **fundamental matrix**.

Fundamental matrix

The point $\mathbf{e}'$ on the epipole in the second view lies on the epipolar line $\mathbf{l}'$.
Therefore,

$$(\mathbf{e}')^\top \mathbf{l}' = (\mathbf{e}')^\top \mathbf{F} \mathbf{p} = 0, \quad \forall \mathbf{p} \in \mathbb{P}^2$$

For this to be true for all $\mathbf{p}$, this implies that $\mathbf{e}'$ lies in the left null space of $\mathbf{F}$.
In other words, the point $\mathbf{e}'$ satisfies the property

$$\mathbf{F}^\top \mathbf{e}' = \mathbf{0}$$

Similarly, the point $\mathbf{e}$ on the epipole in the first view lies on the epipolar line $\mathbf{l} = \mathbf{F}^\top \mathbf{p}'$, which implies

$$\mathbf{l}^\top \mathbf{e} = (\mathbf{p}')^\top \mathbf{F} \mathbf{e} = 0, \quad \forall \mathbf{p}' \in \mathbb{P}^2$$

For this to be true for all $\mathbf{p}'$, this implies that $\mathbf{e}$ lies in the right null space of $\mathbf{F}$,
i.e.,

$$\mathbf{F} \mathbf{e} = \mathbf{0}$$

Fundamental matrix

To find a basis for the left null space of $\mathbf{F}$, let

$$\mathbf{F}\mathbf{\Pi} = \underbrace{[\mathbf{Y} \quad \mathbf{Z}]}_{\mathbf{Q}} \underbrace{\begin{bmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}}_{\mathbf{R}}$$

be a rank-revealing QR decomposition with column pivoting.

Then, the column(s) of $\mathbf{Z}$ form a basis of the left null space of $\mathbf{F}$ since,

$$\mathbf{F}^T \mathbf{Z} = \mathbf{\Pi} \begin{bmatrix} \mathbf{R}_{11}^T & \mathbf{0}^T \\ \mathbf{R}_{12}^T & \mathbf{0}^T \end{bmatrix} \begin{bmatrix} \mathbf{Y}^T \\ \mathbf{Z}^T \end{bmatrix} \mathbf{Z} = \mathbf{\Pi} \begin{bmatrix} \mathbf{R}_{11}^T \\ \mathbf{R}_{12}^T \end{bmatrix} \mathbf{Y}^T \mathbf{Z} = \mathbf{0}$$

Since $\mathbf{F} \in \mathbb{R}^{3 \times 3}$ has rank 2, $\mathbf{Z} \in \mathbb{R}^{3 \times 1}$ and the image of the epipole in the second view is the last column of $\mathbf{Q}$,

$$\mathbf{e}' = \mathbf{Z}$$

Fundamental matrix

To find a basis for the right null space of $\mathbf{F}$, let

$$\mathbf{F}^\top \mathbf{\Pi} = \underbrace{[\mathbf{Y} \quad \mathbf{Z}]}_{\mathbf{Q}} \underbrace{\begin{bmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}}_{\mathbf{R}}$$

be a rank-revealing QR decomposition with column pivoting.

Then, the column(s) of $\mathbf{Z}$ form a basis of the right null space of $\mathbf{F}$ since,

$$\mathbf{F}\mathbf{Z} = \mathbf{\Pi} \begin{bmatrix} \mathbf{R}_{11}^\top & \mathbf{0}^\top \\ \mathbf{R}_{12}^\top & \mathbf{0}^\top \end{bmatrix} \begin{bmatrix} \mathbf{Y}^\top \\ \mathbf{Z}^\top \end{bmatrix} \mathbf{Z} = \mathbf{\Pi} \begin{bmatrix} \mathbf{R}_{11}^\top \\ \mathbf{R}_{12}^\top \end{bmatrix} \mathbf{Y}^\top \mathbf{Z} = \mathbf{0}$$

Since $\mathbf{F} \in \mathbb{R}^{3 \times 3}$ has rank 2, $\mathbf{Z} \in \mathbb{R}^{3 \times 1}$ and the image of the epipole in the first view is the last column of $\mathbf{Q}$,

$$\mathbf{e} = \mathbf{Z}$$

Essential matrix

The fundamental matrix can be viewed as a similarity transformation (that depends on camera parameters) of a purely kinematic transformation,

$$\mathbf{F} = \mathbf{K}^{-\top} \mathbf{E} \mathbf{K}^{-1}$$

where

$$\mathbf{E} = (\mathbf{R}_{c'}^n)^\top \mathbf{S}(\mathbf{r}_{C/N}^n - \mathbf{r}_{C'/N}^n) \mathbf{R}_c^n$$

is known as the **essential matrix**.

Epipolar constraint

Since the point $\mathbf{p}'$ lies on the epipolar line $\mathbf{l}'$, we have

$$(\mathbf{p}')^\top \mathbf{l}' = 0$$

$$(\mathbf{p}')^\top \mathbf{F} \mathbf{p} = 0$$

We refer to this as the **epipolar constraint**.

The epipolar constraint can also be written using the essential matrix

$$\begin{aligned} (\mathbf{p}')^\top \mathbf{F} \mathbf{p} &= (\mathbf{p}')^\top \mathbf{K}^{-\top} \mathbf{E} \mathbf{K}^{-1} \mathbf{p} \\ &= (\mathbf{r}_{P/C'}^c)^\top \mathbf{E} \mathbf{r}_{P/C}^c = 0 \end{aligned}$$

Epipolar geometry in optic flow

Consider two consecutive frames taken from the same camera at time steps $k-1$ and k .

Homogeneous points in the image at time step $k-1$ map onto the epipolar line at time step k ,

$$\mathbf{l}[k] = \mathbf{F}_{k-1}^k \mathbf{p}[k-1]$$

where

$$\mathbf{F}_{k-1}^k = \mathbf{K}^{-\top} (\mathbf{R}_c^n[k])^\top \mathbf{S} (\mathbf{r}_{C/N}^n[k-1] - \mathbf{r}_{C/N}^n[k]) \mathbf{R}_c^n[k-1] \mathbf{K}^{-1}$$

and assuming the environment is static,

$$(\mathbf{p}[k])^\top \mathbf{l}[k] = 0$$

$$(\mathbf{p}[k])^\top \mathbf{F}_{k-1}^k \mathbf{p}[k-1] = 0$$

Epipolar geometry in optic flow

The epipole is the line joins the camera position at time step $k-1$ to the camera position at time step k and passes through the image planes at $\mathbf{e}[k-1]$ and $\mathbf{e}[k]$.

This line represents the direction of translational velocity.

Let

$$\mathbf{F}_{k-1}^k \mathbf{\Pi} = \underbrace{\begin{bmatrix} \mathbf{Y} & \mathbf{Z} \end{bmatrix}}_{\mathbf{Q}} \underbrace{\begin{bmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}}_{\mathbf{R}}$$

be a QR decomposition with column pivoting.

Then, the image of the epipole in frame k is given by

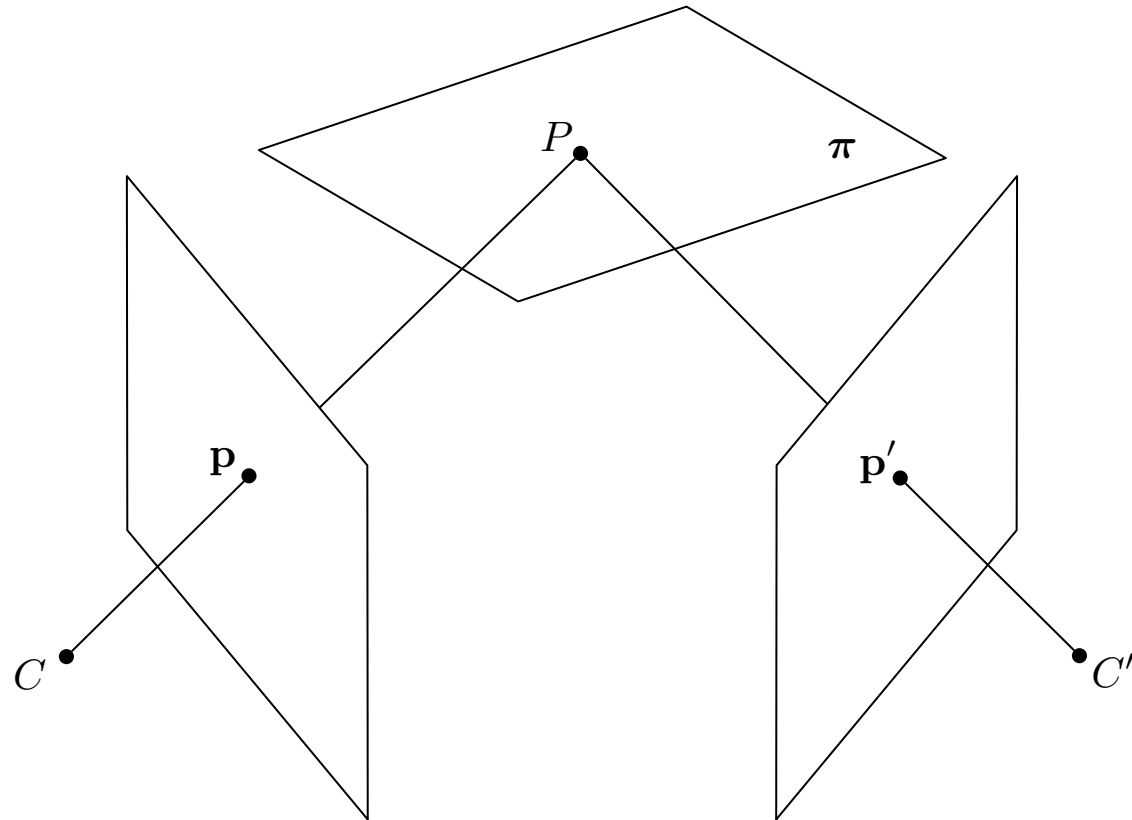
$$\mathbf{e}[k] = \mathbf{Z}$$

World plane homography

Consider two views of the point P , which lies on the plane $\pi \in \mathbb{P}^3$.

Let C and C' be the camera optical centres.

Let $\mathbf{p}, \mathbf{p}' \in \mathbb{P}^2$ be the homogeneous coordinates of the point in each view.



World plane homography

Since the P lies on the plane π ,

$$\begin{aligned}
 0 &= \pi^\top \mathbf{p}_{P/N}^n \\
 &= \pi^\top \mathbf{T}_c^n \begin{bmatrix} \mathbf{K}^{-1} \mathbf{p} \\ \lambda \end{bmatrix} \\
 &= \begin{bmatrix} \mathbf{n} \\ d \end{bmatrix}^\top \begin{bmatrix} \mathbf{R}_c^n & \mathbf{r}_{C/N}^n \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} \mathbf{K}^{-1} \mathbf{p} \\ \lambda \end{bmatrix} \\
 &= \mathbf{n}^\top \mathbf{R}_c^n \mathbf{K}^{-1} \mathbf{p} + \lambda (\mathbf{n}^\top \mathbf{r}_{C/N}^n + d) \\
 \lambda &= -\frac{\mathbf{n}^\top \mathbf{R}_c^n \mathbf{K}^{-1} \mathbf{p}}{\mathbf{n}^\top \mathbf{r}_{C/N}^n + d}
 \end{aligned}$$

and therefore

$$\mathbf{p}_{P/N}^n = \mathbf{T}_c^n \begin{bmatrix} \mathbf{K}^{-1} \mathbf{p} \\ -\frac{\mathbf{n}^\top \mathbf{R}_c^n \mathbf{K}^{-1} \mathbf{p}}{\mathbf{n}^\top \mathbf{r}_{C/N}^n + d} \end{bmatrix} = \mathbf{T}_c^n \begin{bmatrix} \mathbf{I} \\ -\frac{\mathbf{n}^\top \mathbf{R}_c^n}{\mathbf{n}^\top \mathbf{r}_{C/N}^n + d} \end{bmatrix} \mathbf{K}^{-1} \mathbf{p}$$

World plane homography

Projecting onto the second image yields

$$\begin{aligned}
 \mathbf{p}' &= \bar{\mathbf{K}} \mathbf{T}_n^{c'} \mathbf{p}_{P/N}^n \\
 &= \bar{\mathbf{K}} \mathbf{T}_n^{c'} \mathbf{T}_c^n \begin{bmatrix} \mathbf{I} \\ -\frac{\mathbf{n}^\top \mathbf{R}_c^n}{\mathbf{n}^\top \mathbf{r}_{C/N}^n + d} \end{bmatrix} \mathbf{K}^{-1} \mathbf{p} \\
 &= \bar{\mathbf{K}} \mathbf{T}_c^{c'} \begin{bmatrix} \mathbf{I} \\ -\frac{\mathbf{n}^\top \mathbf{R}_c^n}{\mathbf{n}^\top \mathbf{r}_{C/N}^n + d} \end{bmatrix} \mathbf{K}^{-1} \mathbf{p} \\
 &= [\mathbf{K} \quad \mathbf{0}] \begin{bmatrix} \mathbf{R}_c^{c'} & \mathbf{r}_{C/C'}^{c'} \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I} \\ -\frac{\mathbf{n}^\top \mathbf{R}_c^n}{\mathbf{n}^\top \mathbf{r}_{C/N}^n + d} \end{bmatrix} \mathbf{K}^{-1} \mathbf{p} \\
 &= \mathbf{K} \left(\mathbf{R}_c^{c'} - \frac{\mathbf{r}_{C/C'}^{c'} \mathbf{n}^\top \mathbf{R}_c^n}{\mathbf{n}^\top \mathbf{r}_{C/N}^n + d} \right) \mathbf{K}^{-1} \mathbf{p}
 \end{aligned}$$

World plane homography

Therefore, we have the following homography between $\mathbf{p}$ and $\mathbf{p}'$:

$$\mathbf{p}' = \mathbf{H}_\pi \mathbf{p}$$

where

$$\begin{aligned} \mathbf{H}_\pi &= \mathbf{K} \left(\mathbf{R}_c^{c'} - \frac{\mathbf{r}_{C/C'}^{c'} \mathbf{n}^\top \mathbf{R}_c^n}{\mathbf{n}^\top \mathbf{r}_{C/N}^n + d} \right) \mathbf{K}^{-1} \\ &= \mathbf{K} \left((\mathbf{R}_{c'}^n)^\top \mathbf{R}_c^n - \frac{(\mathbf{R}_{c'}^n)^\top (\mathbf{r}_{C/N}^n - \mathbf{r}_{C'/N}^n) \mathbf{n}^\top \mathbf{R}_c^n}{\mathbf{n}^\top \mathbf{r}_{C/N}^n + d} \right) \mathbf{K}^{-1} \\ &= \mathbf{K} (\mathbf{R}_{c'}^n)^\top \left(\mathbf{I} - \frac{(\mathbf{r}_{C/N}^n - \mathbf{r}_{C'/N}^n) \mathbf{n}^\top}{\mathbf{n}^\top \mathbf{r}_{C/N}^n + d} \right) \mathbf{R}_c^n \mathbf{K}^{-1} \end{aligned}$$

World plane homography

There are two planes of particular interest in outdoor environments:

- The ground plane ($z = 0$),

$$\boldsymbol{\pi}_z = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \implies \mathbf{H}_z = \mathbf{K}(\mathbf{R}_{c'}^n)^\top \left(\mathbf{I} - \frac{(\mathbf{r}_{C/N}^n - \mathbf{r}_{C'/N}^n)\mathbf{e}_3^\top}{\mathbf{e}_3^\top \mathbf{r}_{C/N}^n} \right) \mathbf{R}_c^n \mathbf{K}^{-1}$$

- The plane at infinity ($w = 0$),

$$\boldsymbol{\pi}_\infty = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \implies \mathbf{H}_\infty = \mathbf{K}(\mathbf{R}_{c'}^n)^\top \mathbf{R}_c^n \mathbf{K}^{-1}$$

World plane homography in optic flow

Consider two consecutive frames taken from the same camera at time steps $k-1$ and k .

If these frames contain homogeneous points that image the same world point, and that world point is known to coincide with a given plane, then

$$\mathbf{p}[k] = \mathbf{H}_{k-1}^k \mathbf{p}[k-1]$$

where for the ground plane,

$$\mathbf{H}_{k-1}^k = \mathbf{K}(\mathbf{R}_c^n[k])^\top \left(\mathbf{I} - \frac{(\mathbf{r}_{C/N}^n[k-1] - \mathbf{r}_{C/N}^n[k])\mathbf{e}_3^\top}{\mathbf{e}_3^\top \mathbf{r}_{C/N}^n[k-1]} \right) \mathbf{R}_c^n[k-1] \mathbf{K}^{-1}$$

and for the skydome,

$$\mathbf{H}_{k-1}^k = \mathbf{K}(\mathbf{R}_c^n[k])^\top \mathbf{R}_c^n[k-1] \mathbf{K}^{-1}$$

References

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